Venkata Lakshmi Parimala Pasupuleti

Forward Deployed Engineer · GenAI & Agentic Platforms · LLMOps · Enterprise AI · Python · AWS

LinkedIn | GitHub | Portfolio | Virginia, USA

PROFESSIONAL SUMMARY

AI/ML Engineer with 8+ years shipping production systems and hands-on GenAI development. Built AgentProbe - a 3-agent LangGraph agentic platform with a 5-metric LLM evaluation engine, multi-provider routing (Anthropic, OpenAI, Groq, Ollama), and CI/CD-integrated LLMOps pipeline. Built LeaseI - an enterprise AI RAG system achieving 93.6% precision with deterministic checks and fully auditable outputs. At Entrata, shipped the company's first GenAI pipeline on 1.6M financial records and translated business requirements into 20+ production modules for 100K+ MAU. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026). Experienced in rapid prototyping of agentic platforms, enterprise system integration, and mentoring (600+ students, 5+ engineers).

CORE COMPETENCIES

Generative AI (GenAI) | Agentic Platforms | LLM Solutions | LLMOps / MLOps | RAG Pipelines | LLM Evaluation | Rapid Prototyping | Enterprise AI | Multi-Agent Orchestration | Data Engineering | Client Collaboration | AWS Cloud | Enterprise System Integration | Security & Compliance AI

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer Intern - GenAI Agentic Platform · Rapid Prototyping · Enterprise AI Delivery

- Rapidly prototyped and deployed a **GenAI agentic platform** (LangGraph stateful graph: parse → plan → retrieve → validate → route) - translated business requirements from marketing, real estate, and commercial survey stakeholders into a production-ready AI implementation.
- Built a **RAG layer** over FAA Part 107 compliance corpus with cited-source grounding; engineered validation guardrails enforcing altitude limits, battery budgets, and no-fly-zone checks - enterprise AI safety controls before export.

George Mason University - College of Engineering and Computing

Aug 2024 - May 2026

Graduate Teaching Assistant · Research Associate - Mentorship · Data Engineering · Statistical Modeling

- Mentored **600+ students** across MIS 303 sections - Python, data structures, SQL, ER modeling, database normalization; one-on-one debugging support and supplementary exercise development.
- Built multi-source prediction models (Python, R) on WHO + UK Road Safety datasets across 190+ countries - $R^2 = 0.977$; applied Isolation Forest to automate data integrity checks across 3 research pipelines; designed Tableau/ggplot2 dashboards improving comprehension 40%.

Entrata - PropTech SaaS (100K+ MAU, Enterprise Scale)

Sep 2018 - Aug 2024

Senior Software Engineer - GenAI Pipeline · Enterprise System Integration · Data Engineering · AWS · Production AI

- **Shipped Entrata's first production GenAI pipeline** on 1.6M financial records (Python, PyTorch, OpenAI API) - translated business requirements into an enterprise AI implementation with quality gates and compliance controls; zero incidents at launch.
- **Built 20+ full-stack production modules** (Python, PHP, PostgreSQL, RabbitMQ) across 120+ Agile sprints for 100K+ MAU - 99.5% uptime; collaborated with compliance, operations, and product stakeholders translating requirements into scalable enterprise systems.
- **Data engineering at scale:** ETL pipelines and schema design for 1.6M+ records across income eligibility, HUD payments, and financial reporting; optimized 200+ SQL queries (30% gain); AWS S3/Glue/RDS/EC2 infrastructure with 100% PHI compliance.
- **Enterprise system integration:** Docker microservices redesign (50% scalability improvement); Salesforce + n8n CRM data pipelines; mentored 5+ engineers across 300+ peer code reviews; eliminated 80% manual workload via Python/Bash automation.

Entrata India (Db Xento Systems)

Jan 2017 - Sep 2018

Associate / Trainee Software Engineer - Full-Stack · Data Pipelines · AWS S3

- Built property management modules (PHP, Python, PostgreSQL, JS); SQL import scripts for 1.6M HUD/Tax Credit records; migrated client documents to AWS S3 resolving 90% of storage constraints.

PROJECTS

AgentProbe: Agentic AI Testing Toolkit

Agentic Platform | LLMOps | LLM Evaluation

3-agent LangGraph agentic platform (Orchestrator, RAG Tracer, Auto Fix) with multi-provider LLM routing (Anthropic, OpenAI, Groq, Ollama), 15-endpoint FastAPI backend, and 5-metric LLM-as-judge eval engine integrated into CI/CD - failing runs blocked automatically. React observability dashboard tracks per-run regression drift and retrieval quality.

Python, LangGraph, FastAPI, ChromaDB, Anthropic, OpenAI, React, Docker, GitHub Actions · github.com/venkatalakshmiparimala/AgentProbe

Leasel: Enterprise AI Compliance System

Enterprise AI | RAG Pipeline | Deterministic Checks

93.6% precision on 1,240 lease PDFs - agentic GenAI RAG pipeline with deterministic rules engine for fully auditable enterprise AI outputs, LLM-as-judge scoring, pass@k harness (pass@1 81.7%, pass@3 88.3%). Every compliance determination traceable to cited source.

Python, LangGraph, SQLite · github.com/venkatalakshmiparimala/Lease_complaince_rag

AI on FHIR: NLP-Driven Medical Records Query System

Client-Facing AI | Rapid Prototyping | HIPAA Compliance

Rapidly prototyped and deployed a GenAI platform translating natural-language queries (EN/ES) to FHIR APIs using LLMs, BERT, and embeddings -- 60% reduction in clinical retrieval time, HIPAA-compliant RBAC, enterprise system integration with FHIR standards.

Flask, LLMs, FHIR, React, Docker, spaCy, BERT · github.com/venkatalakshmiparimala/AI_on_FHIR

EDUCATION

M.S. Data Analytics Engineering - George Mason University — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTUA — 2017

TECHNICAL SKILLS

GenAI & Agentic: LangGraph, LangChain, Multi-Agent Orchestration, Agentic Platforms, Tool Calling, LLM Solutions, GenAI Pipelines, Rapid Prototyping, OpenAI API, Anthropic API, Groq, Ollama, Prompt Engineering, HITL

LLMOps / MLOps & Evaluation: LLM Evaluation, LLM-as-Judge, RAG Pipelines, ChromaDB, Vector Database, Embeddings, Observability, CI/CD Integration, FastAPI, REST APIs, PyTorch, scikit-learn, XGBoost

Data Engineering & Enterprise Systems: Python, PHP, SQL, ETL/ELT, Schema Design, Data Normalization, AWS (S3, EC2, RDS, Glue), Docker, PostgreSQL, MySQL, Enterprise System Integration, Microservices, RabbitMQ

Security, Compliance & Client Collaboration: PHI Compliance, HIPAA-Compliant AI, Deterministic Validation Gates, Audit Pipelines, Agile/Scrum (120+ sprints), Cross-Functional Stakeholder Collaboration, Mentorship, Technical Documentation